The examination is being carried out on the **following application documents**

Description, Pages

1-35 as published

Sequence listings, SEQ ID NO

1-42 as published

Claims, Numbers

1-15 filed in electronic form on 25-09-2023

Drawings, Sheets

1/5-5/5 as published

1 The amendments filed with the letter of 25.09.2023 meet the requirements as set forth in Article 123(2) EPC.

2 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 WO 2016/016871 A1 (FUNDACIÓ HOSPITAL UNI VALL D'HEBRON INST DE RECERCA [ES]) 4 February 2016 (2016-02-04)

D2 LING STEPHANIE F ET AL: "Proteomic analysis to define predictors of treatment response to adalimumab or methotrexate in rheumatoid arthritis patients",THE PHARMACOGENOMICS JOURNAL, NATURE PUBLISHING GROUP, GB, vol. 20, no. 3, 10 December 2019 (2019-12-10), pages 516-523, XP037147851, ISSN: 1470-269X, DOI: 10.1038/S41397-019-0139-4

The following documents have been cited in the international search report; the numbering will be adhered to in the rest of the procedure.

D3 .: "IMMUNOME TM",Houston Omics Collaborative, 4 July 2018 (2018-07-04), pages 1-2, XP055940928,

D4 ORTEA I. et al.: "Independent Candidate Serum Protein Biomarkers of Response to Adalimumab and to Infliximab in Rheumatoid Arthritis: An Exploratory Study",PLoS One, vol. 11, no. 4 6 April 2016 (2016-04-06), page e0153140, XP055890906,DOI: 10.1371/JOURNAL.PONE.0153140

D5 Matsudaira Ran, Tamura Naoto, Sekiya Fumio, Ogasawara Michihiro, Yamanaka Kenjiro, Takasaki Yoshinari: "Anti-Ro/SSA Antibodies Are an Independent Factor Associated with an Insufficient Response to Tumor Necrosis Factor Inhibitors in Patients with Rheumatoid Arthritis", JOURNAL OF RHEUMATOLOGY, Journal of Rheumatology Publishing Company., CA, vol. 38, no. 11, 1 November 2011 (2011-11-01), pages 2346-2354, XP055940932,CA DOI: 10.3899/jrheum.101295

D6 WO 03/064656 A1 (SENSE PROTEOMIC LIMITED) 7 August 2003 (2003-08-07)

D7 Chen Po-Ku, Lan Joung-Liang, Chen Yi-Ming, Chen Hsin-Hua, Chang Shih-Hsin, Chung Chia-Min, Rutt Nurul H., Tan Ti-Myen, Mamat Raja: "Anti-TROVE2 Antibody Determined by Immune-Related Array May Serve as a Predictive Marker for Adalimumab Immunogenicity and Effectiveness in RA",Journal of Immunology Research, Hindawi Publishing Corporation, US, vol. 2021, 8 March 2021 (2021-03-08), pages 1-13, XP055940937,US DOI: 10.1155/2021/6656121

D8 Mori Ayano, Saito Toshiharu, Takahashi Miho, Shibata Miho, Tsuji Goh, Hatachi Saori, Takahashi Soshi, Kumagai Shunichi: "Presence of anti-nuclear antibodies is a risk factor for the appearance of anti-drug antibodies during infliximab or adalimumab therapy in patients with rheumatoid arthritis", PLOS ONE, vol. 15, no. 12 , page e0243729, XP055940942, DOI: 10.1371/journal.pone.0243729

3 Documents D1 and D2 supra teach predictors of treatment response to adalimumab in rheumatoid arthritis patients. D1 and D2 do not teach the markers defined in present claims.

4 An International Preliminary Report on Patentability has already been drawn up for the present application in accordance with the PCT by the Singapore Intellectual Property Office.

The present search authority under the EPC agrees with the Report above.

4.1 P-Documents D7 and D8 are not relevant for the assessment of novelty and inventive step, because the priority of the application is validly claimed.

4.2 None of the documents D1 to D6 disclose the present methods and uses based on the determination of autoantibodies to antigens SSB, TROVE2 and ZHX2.

4.3 Each of D1, D2, D4 or D5 can be considered the most relevant prior art with respect to present claims 1-15. The technical problem to be solved by said claims is the provision of a panel of autoantibody biomarkers that are able to predict immunogenicity of adalimumab and therapeutic responses to adalimumab in individual RA patients, prior to treatment with adalimumab.

The known prior art does not suggest the present methods and uses. Claims 1-15 meet the requirements of Articles 54 and 56 EPC.

5 Claim 9 does not meet the requirements of Article 84 EPC in that its subject-matter is not clearly defined: it pertains to the use of a kit to predict a response to adalimumab, however the components of the kit are not defined, and the steps of manufacturing the kit are included in the claim, rendering the scope and category of the claim unclear.

6 The applicant is invited to file new claims which take account of the above comments.

7 Care should be taken during revision, especially of the introductory portion and of any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

8 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages **of the application as originally filed** on which these amendments are based (see Guidelines H III, 2.1).